Assignment 4: Implement Berkeley algorithm for clock synchronization

Step 1: Create New Notebook

- Click **New** file
- berkeley.py

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Viva Questions and Answers for Berkeley Algorithm

1. What is Berkeley Algorithm?

Berkeley Algorithm is a clock synchronization algorithm used in distributed systems.

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2. What is clock synchronization?

Clock synchronization means making all system clocks show approximately the same time.

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3. Why is clock synchronization needed?

It helps maintain proper coordination and consistency in distributed systems.

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4. Who developed Berkeley Algorithm?

It was developed at:

University of California, Berkeley

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5. Is Berkeley Algorithm centralized or decentralized?

Centralized

Because one master system controls synchronization.

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6. What is the role of the master clock?

The master clock:

- Collects time from all systems
- Calculates average time

- Sends corrected time to all systems

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7. Does Berkeley Algorithm use UTC time?

No

It uses the average system time.

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8. What is average time in Berkeley Algorithm?

Average of all processor/system clock values.

Example:

(10 + 15 + 20 + 25) / 4 = 17.5

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9. What happens after calculating average time?

All processor clocks are updated to the average time.

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10. What is distributed system?

A distributed system is a collection of multiple computers working together.

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11. What is clock drift?

Clock drift means clocks run at slightly different speeds over time.

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12. Advantages of Berkeley Algorithm

- Simple algorithm
- Better synchronization
- No need for external UTC clock

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13. Disadvantages of Berkeley Algorithm

- Master failure affects synchronization
- Network delays reduce accuracy

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14. What is the output of your program?

Clock Times Before Synchronization:
[10, 15, 20, 25]

Average Time = 17.5

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15. Which language is used in your program?

Python

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16. Which platform is used?

Jupyter Notebook

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17. What is the use of sum() function?

It calculates the total of all clock values.

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18. What does len(clocks) return?

It returns the number of clocks/processors.

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19. Why is average calculated?

To synchronize all clocks to a common time.

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20. Real-life applications of clock synchronization

- Banking systems
- Distributed databases
- Cloud computing
- Airline reservation systems